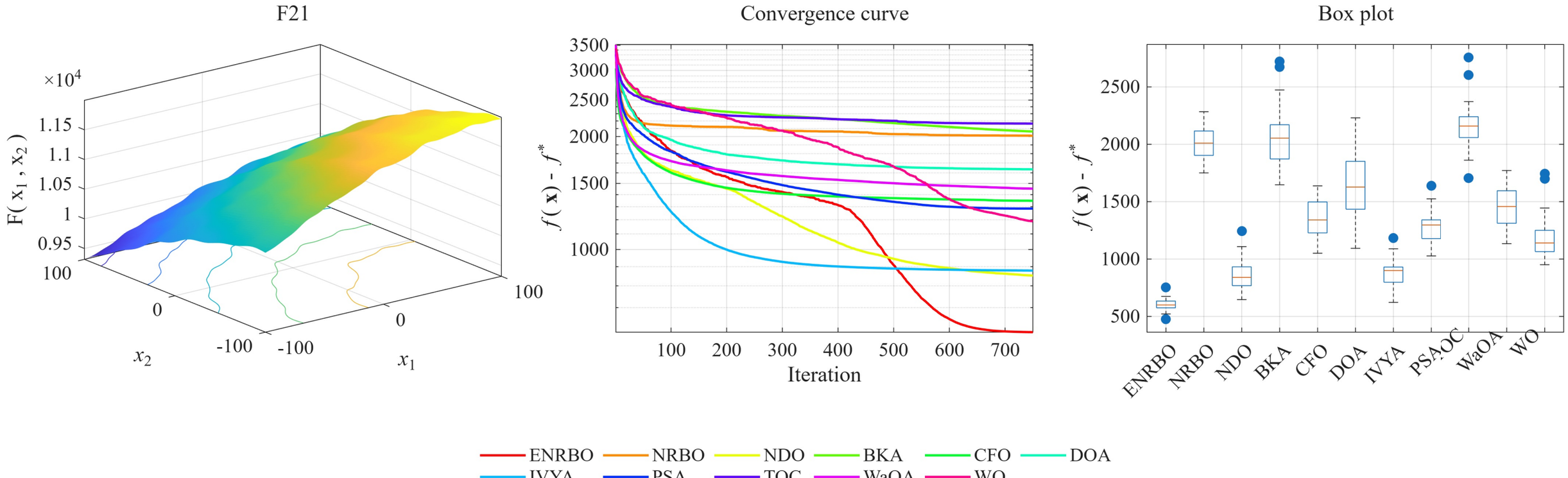

F21
×10^4
1.15
1.1
1.05
1
0.95
F( x1 , x2 )
100
0
-100
x2
-100
0
100
x1
Convergence curve
3500
3000
2500
2000
1500
1000
f( x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
2500
2000
1500
1000
500
f( x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO